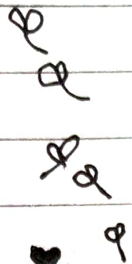


Actualisation



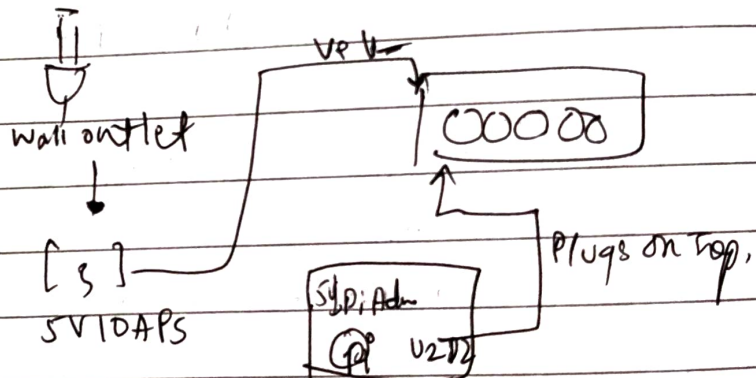
Research and Industrial Robotics

- ↳ Designed for multi joint robot
- ↳ Position, load, temp., voltage, feedback.

(Daisy chain)

Snake Robot Require highly coordinated, multi-sensor synchronization.

2 Power Source (Mains)



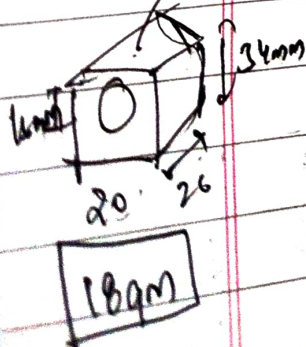
*Dynamixel XL330

5mps

V2D1 $\longleftrightarrow$ V2D2 Power humb-
connect.

~~At~~ 12V, 5 Amp

to Dynamical.



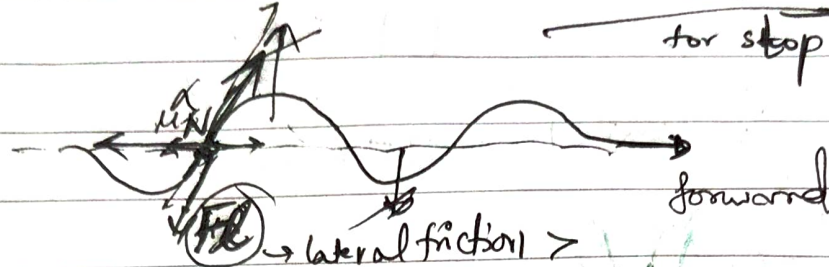
That how you power dynamical using $U_2 D_2$

→ lateral Undulation (serpentine crawling or horizontal slithering)
 → Anisotropic friction (meaning friction is much higher in one direction)
Maths & Physics

Anisotropic friction

$$\Sigma F_{\text{lateral}} = 0$$

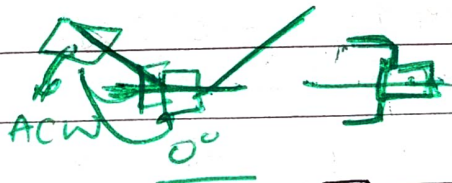
for stop drifting side to side



S shape allows the Robot to Generate a normal force.

Snake head moves (it does not make complete S shape by itself because no motor ahead of it, so the head wags left and right like a pointer).

→ we can also make head stable (and make 2 to 10 motor do all heavy lifting and wiggling behind it to push the Robot forward).

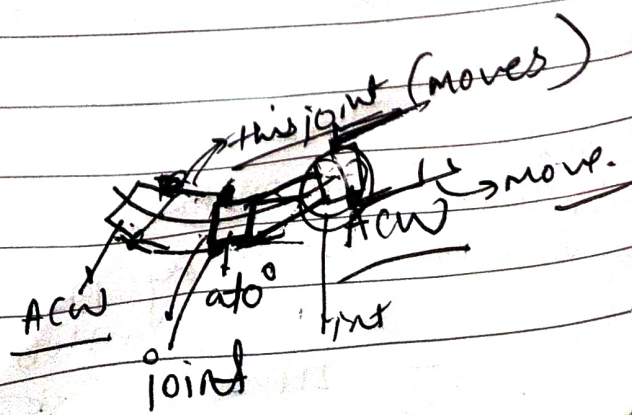


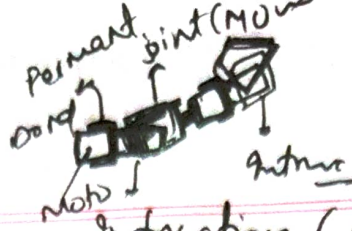
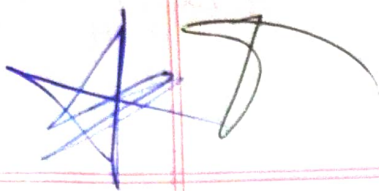
Powered in parallel for High-Current Undulation.

(Python Wave Math in pi Control & Articulation.

Head

what happened

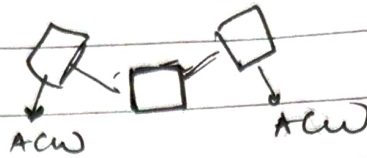




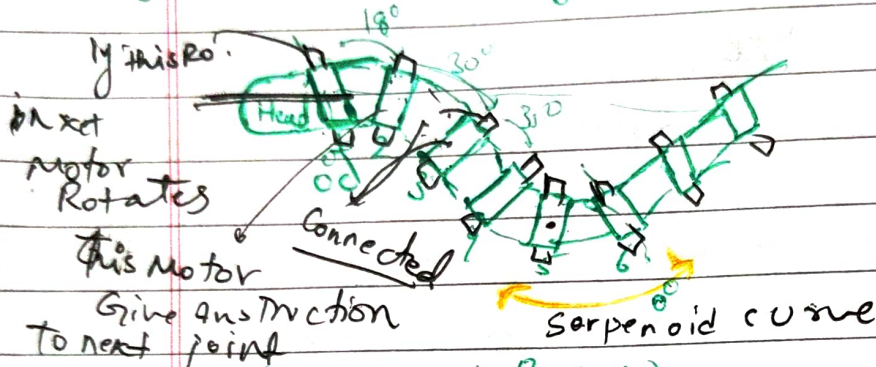
classmate

Date _____
Page _____

Instruction (ACW 35°)



(CW (+ve angle) → left bend



Motor 2 = $+18^\circ$ CCW

Motor 3 = $+30^\circ$ CCW

Motor 4 = $+35^\circ$ CCW

Motor 5 = 18° CCW

Motor 6 = 0° (straight)

Motor 7 = -18° CW

Motor 8 = -30° CW

Motor 9 = -35° CW

Motor 10 = -18° CW

Each segment generate its own lateral Force. (sideways) when motor pushes against the ground (if no tyre).

All on Robotic snake for complete autonomy

Total mass = $189\text{gm} \times 10$

3D-printed link bracket = $129\text{gm} \times 10$

Raspberry pi = 1 = $1 \times 469\text{gm} = 469\text{gm}$

U2D2+ Power Hub Board = 259gm

LiPo Battery Pack = 649gm

Total around mass = 4609gm

⇒ We need material which provide *classmate*
max anisotropic friction

Date _____
Page _____

on snake body

→ On Motor 1 (head segment) = Sensor.

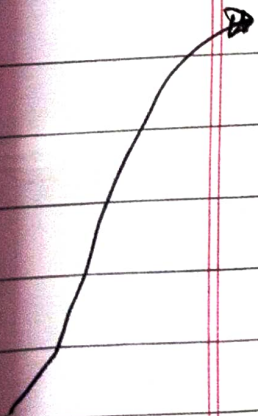
→ Front Body (2, 3 & 4) = U2D2 + Power Hub Board.
(on front) because they act as wiring hub where the data from the computer & power from the battery split up).

→ Center Body :- (segments 5, 6 & 7) ⇒ Hold P_i

→ Tail End (8 & 9 & 10) = LiPo Battery Pack.

★ every ~~one~~ single wire is going to be flexing back & forth by $\pm 35^\circ$ continuous loop.

lateral Force, (Peak segment apply max. lateral Force).



Material to use for Anisotropic friction

→ Directional Velvet fabric

↳ work because of unique texture.

Tangential → million of tiny fabric fold completely flat.
→ it behave like a smooth silk ribbon, sliding over the tiles with a very low fric. ($\mu \approx 0.05$)

→ Lateral dirⁿ ⇒ when dynamical motor twist the joint sideways, the fabric forced against the grain. → then microhalis instant stiffen; stand up and act like millions of miniature hooks → catch catching inside those microscopic pits on the floor. $\mu \approx 0.85$

→ By This Material Calculation.

$$M_{\text{ars}} = 460 \text{ gm} \approx 500 \text{ gm}$$

$$N = 0.5 \times 9.81 = 4.95 \text{ N}$$

$$N_{\text{segment}} = 0.49 \text{ N}$$

→ For lateral Force.

$$F_{\text{lateral}} = \mu_N \times N_{\text{segment}}$$

$$F_{\text{lateral}} = 0.85 \times 0.49 = 0.416 \text{ N}$$

$$\rightarrow F_{\text{drag}} (\text{Forward Drag Force}) = 0.15 \times 0.49 \text{ N} \\ = 0.07 \text{ N}$$

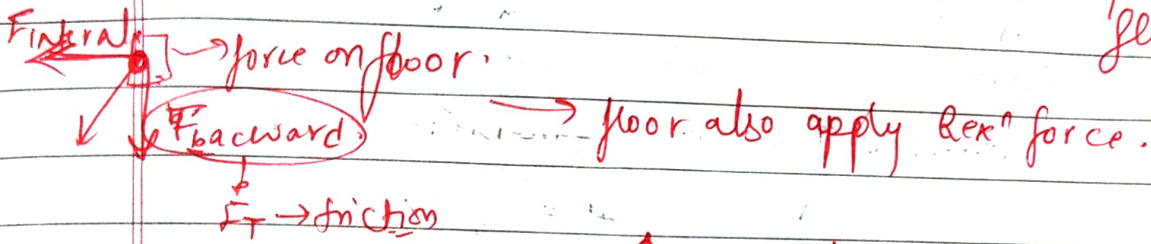
Snake robot Internal Rotational work $\rightarrow$ Linear Translation force.
convert



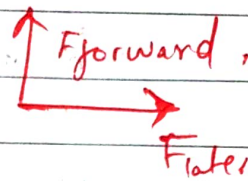
Motor housing is bolted to link i and its rotating output shaft is bolted to link $(i+1)$

Anchor (Temp. rock solid point of contact)

Motor apply torque to swing the joint, link $i+1$ is physically forced to push sideways against the floor.



① Floor can Resist upto



② Floor push back against the link.

③ Floor hold the side of the wedge firmly in place with an anchoring force of 0.416 N at pivot point

so this Give Motor a rigid wall push off against.

* F_{drag} :- As the Motor pushes Against lateral anchor. The segment is forced to slide forward along its own spine line.

$\rightarrow$ To move Forward, Forward Component of anchoring force must overcome This drag (backward force)

Net Thrust per segment $\Rightarrow F_{\text{lateral}} \sin(\theta) - F_{\text{drag}} \cos(\theta)$
simply (for that motor)

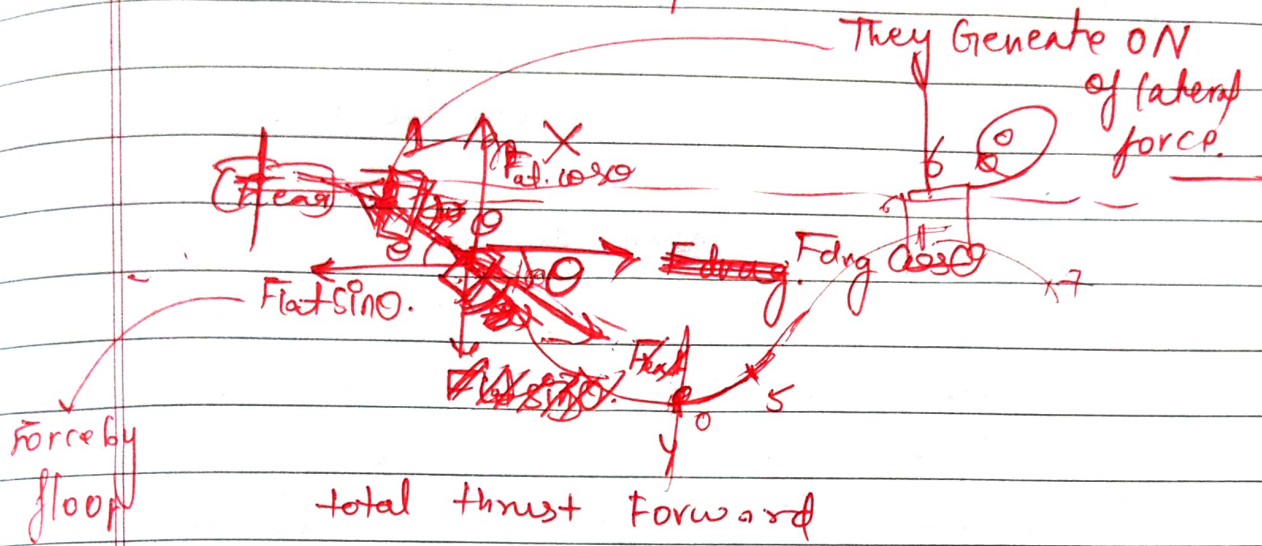
$\Sigma F_{lateral} = 0$ (We have 10 motors)

$F_{lat} \sin \theta$

F_{drag}

F_{lat}

→ System adds All the individual segment Contributions together.



$\Sigma F_{thrust} = 6 \text{ active joint} \times 0.12 \text{ (resolved thrust)}$

$\approx 0.72 N$

→ Two zones of S-Wave.

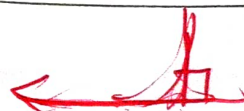
① The slanted Slope (Active Thrusters)

2, 3, 5, 7, 8.

These are Angled sharply ($\theta = 15^\circ$ to 30°) Relative to your forward tracking line

→ they generate Thrust because motor torque forces push them backward and outward.

② 4 and 9 are peak
 $\sin \theta = 0$



Torque required

$$\rightarrow \text{Mass} = 500 \text{ gm} \\ \text{per segment} = 50 \text{ gm} = 0.5 \text{ kg}$$

$$\text{Normal, } N_{\text{segm}} = m_{\text{seg}} \times g \approx 0.49 \text{ N}$$

$$\rightarrow \text{Maxi lateral friction force} \rightarrow 0.85 \times 0.49 \text{ N} \\ = 0.4165 \text{ N lat.}$$

$$\rightarrow \tau = F \times L \times \sin \theta$$

assuming distance b/w joint center 40 mm

$$= 0.04 \text{ m}$$

$$\rightarrow \text{Maxi. Torque}$$

$$\tau = F_{\text{lat}} \times L \times \sin(35^\circ)$$

$$\tau = 0.0095 \text{ N-m}$$

$\rightarrow$ for safety purpose: (for back & forth, drag small wire)

$$\tau_{\text{req.}} = 2 \times 0.0095 \text{ N-m} = 0.019 \text{ N-m}$$

$$\text{Stall Torque at S.V} = 0.52 \text{ N-m}$$

$$0.15 \text{ N-m}$$

$$\tau_{\text{required}} (0.019) \ll \tau_{\text{working limit}} (0.15 \text{ N-m})$$

only less than 13%

m code implem

$$\theta_i(t) = \alpha \sin(\omega t + i\beta)$$

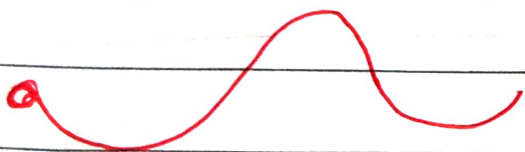
$$\frac{M_n}{M_t} = \frac{0.85}{0.15} = 5.67$$

$$\rightarrow \phi_i(t) = 2\alpha \sin\left(\frac{\beta}{2}\right) \sin(\omega t + i\beta)$$

Maxi wing angle = (without self-collision)
is b/w 30° and 45°

For 10 motor snake to display roughly 1.5 full waves.

B = Spatial wave free



$$\beta \approx \frac{3\pi}{\text{num_motors}}$$

$$\frac{1.5 \times 360}{10} = 54^\circ = \beta$$

FRICTION RESOLUTION FOR FORWARD PROPULSION IN A SEIGMENTED SNAKE ROBOT.

